

NANYANG TECHNOLOGICAL UNIVERSITY

Lecture 9

EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

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Learning Objectives

By the end of this lecture, you should be able to:

- Analyse the DC motor using the magnetisation curve, taking into account the armature reaction effects.
- Describe the various methods of speed control and the speed control limits of a DC motor.
- Explain the techniques that may be used for starting a DC motor.

Nonlinear Analysis of a Shunt DC Motor

- The flux, and hence, the internal generated voltage of a DC machine is a non-linear function of its mmf.
- Anything that changes mmf will have a nonlinear effect on the internal generated voltage.
- Hence, an accurate calculation of the motor torque, the internal generated voltage and speed of the motor for various operating conditions require the use of the magnetisation curve.

Nonlinear Analysis of a Shunt DC Motor

- The two principal contributions to the mmf are its field current and armature reaction, if present.
- The magnetisation curve of the machine is used to determine the generated voltage for a given mmf or equivalently, the field current.
- If the machine has armature reaction, the total net mmf:

$$mmf_{net} = N_F I_F - mmf_{AR}$$

Letting, $mmf_{net} = N_F I_F^*$

where I_F^* = equivalent field current that will produce the same output voltage as the combination of all the mmfs in the machine. Then $I_F^* = I_F - \frac{mmf_{AR}}{N_F}$.

Nonlinear Analysis of a Shunt DC Motor

- The resulting E_A can then be determined by locating I_F^* on the magnetisation curve.

- The magnetisation curve for a machine is drawn for a particular speed, usually the rated speed. How can the effects of a given field current be determined, if the motor is turning at other than rated speed?

- For a given effective field current, the flux is fixed. Then,

$$E_A = \hat{K}\phi n_m \Rightarrow \frac{E_A}{E_{A0}} = \frac{n_m}{n_0}$$

where E_{A0} and n_0 represent the reference values of voltages and speed respectively.
n_m is the actual speed that the motor is turning.

Example 1

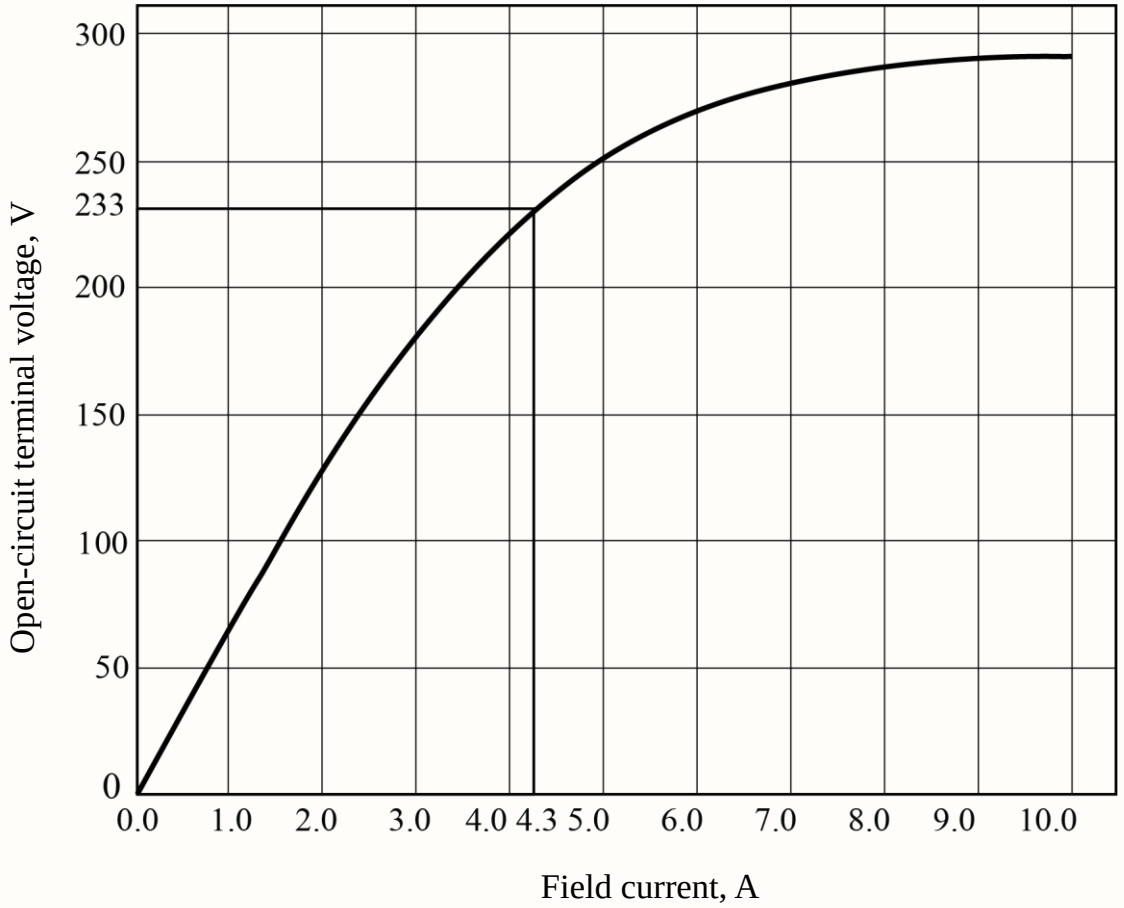
A 50-hp, 250-V DC shunt motor without compensating windings has an armature resistance (including the brushes and interpoles) of 0.06 ohms. Its field circuit has a total resistance of 50 ohms. Assume that the no-load speed is 1200 r/min. There are 1200 turns per pole on the field winding, and the armature reaction produces a demagnetising mmf of 840 A-turns when the input current is 200 A. The magnetisation curve of this machine is as shown.

a) Find the speed of the motor when its input current is 200 A.

b) How does its speed compare to that of the motor in the previous example at an input current of 200 A?

(Solutions →)

Example 1



The magnetisation curve of a typical 250-V DC motor, at 1200 r/min.

Example 1 – Solutions

a) If $I_L = 200$ A, $I_A = I_L - I_F = 195$ A, $E_A = V_T - I_A R_A = 238.3$ V.

At $I_L = 200$ A, the demagnetising mmf due to armature reaction is 840 A-turns.

The effective field current: $I_F^* = I_F - \frac{mmf_{AR}}{N_F} = 5 - \frac{840}{1200} = 4.3$ A

From the magnetisation curve, it produces an E_{A0} of 233 V at n_0 of 1200 r/min.

The actual generated voltage is 238.3 V. Therefore,

$$\frac{E_A}{E_{A0}} = \frac{n_m}{n_0} \Rightarrow \frac{238.3}{233} = \frac{n_m}{1200} \Rightarrow n_m = 1227 \text{ r/min}$$

Example 1 – Solutions

b) The motor speed at an input current of 200 A in the previous example was 1144 r/min. In this example, the motor's speed is 1227 r/min. Notice that the speed of the motor with armature reaction is higher than the speed of the motor with no armature reaction. This relative increase in speed is due to the flux weakening effect in the machine with armature reaction.

Speed Control of a Shunt DC Motor

- The torque-speed equation is:

$$\omega_m = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2} T_{ind}$$

- The speed of a DC shunt motor can be controlled by:
  - Adjusting the field resistance (and thus the field flux).
  - Adjusting the terminal voltage applied to the armature.
  - Inserting a resistor in series with the armature circuit.

Speed Control of a Shunt DC Motor

- Adjusting the field resistance
  - Increasing R_F leads to,

$$I_F = \frac{V_T}{R_F} \downarrow \Rightarrow \phi \downarrow \Rightarrow E_A = K\phi\omega_m \downarrow \Rightarrow I_A = \frac{V_T - E_A}{R_A} \uparrow$$

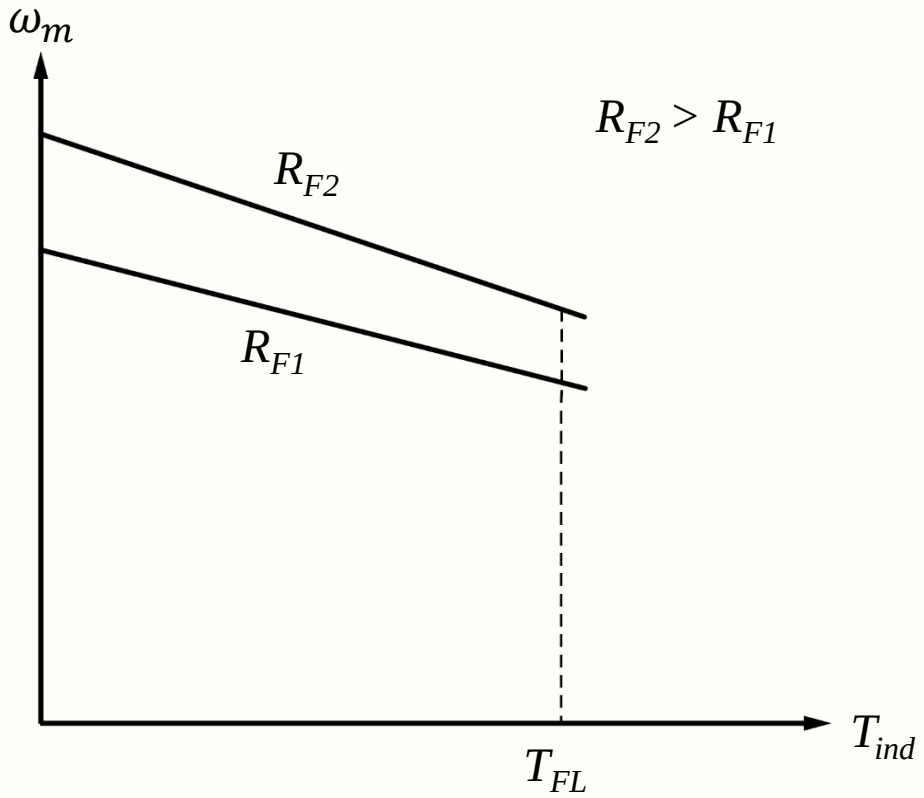
$\Rightarrow T_{ind} = K\left(\phi \downarrow I_A \Uparrow\right)$ increases, with the change in I_A dominant over the change in flux.

$\Rightarrow T_{ind} > T_{load} \Rightarrow \omega_m \uparrow \Rightarrow E_A = K\phi\omega_m \uparrow$ again

$\Rightarrow I_A \downarrow \Rightarrow T_{ind} \downarrow$ until $T_{ind} = T_{load}$ at a higher steady-state speed than originally.

Speed Control of a Shunt DC Motor

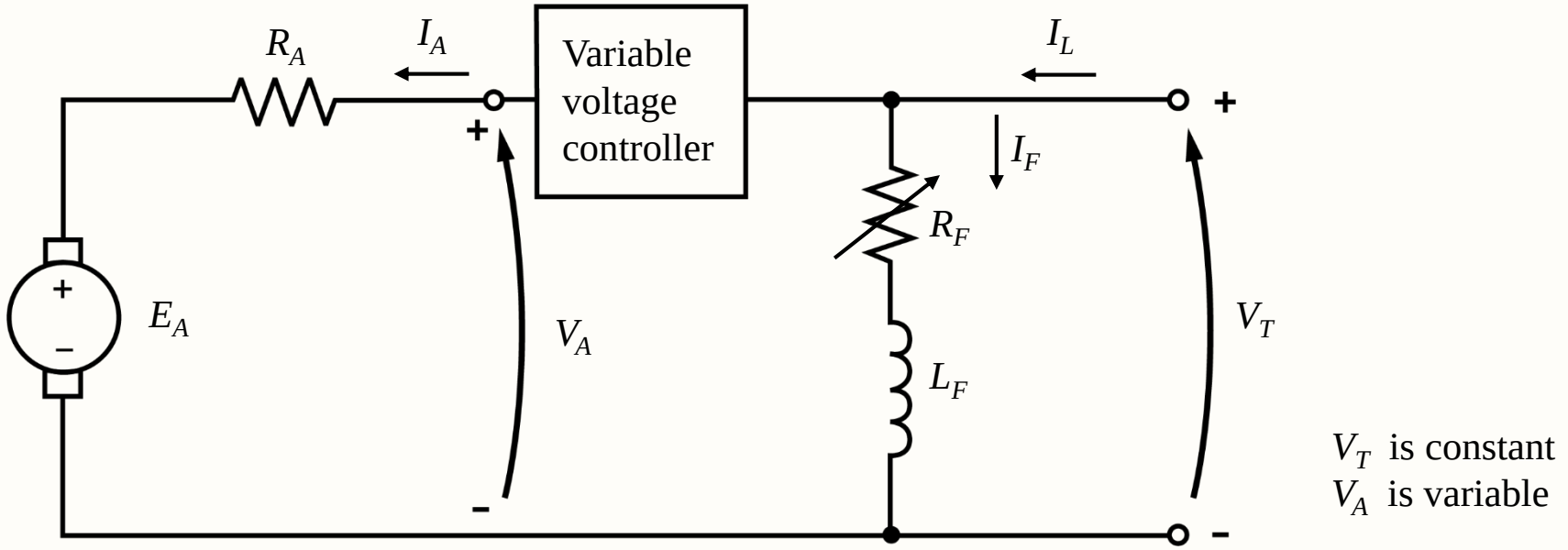
- The effect of field resistance speed control on a shunt motor's torque-speed characteristics.



Over the normal motor's operating range.

Speed Control of a Shunt DC Motor

- Adjusting the armature terminal voltage
  - Change the voltage applied to the armature, without changing the voltage applied to the field.
  - The motor must be separately excited to use armature voltage control.

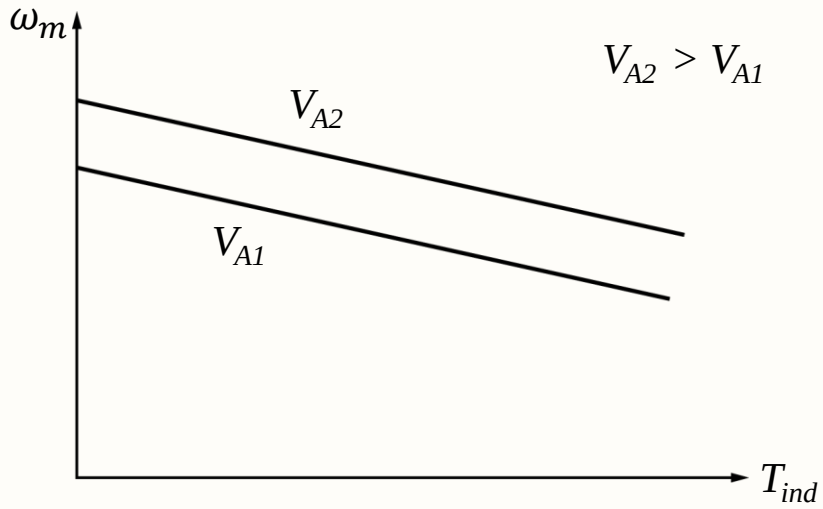


Speed Control of a Shunt DC Motor

- Increasing V_A leads to,

$$I_A = \frac{V_A - E_A}{R_A} \uparrow \Rightarrow T_{ind} = K\phi I_A \uparrow \Rightarrow T_{ind} > T_{load} \Rightarrow \omega_m \uparrow \Rightarrow E_A = K\phi\omega_m \uparrow$$

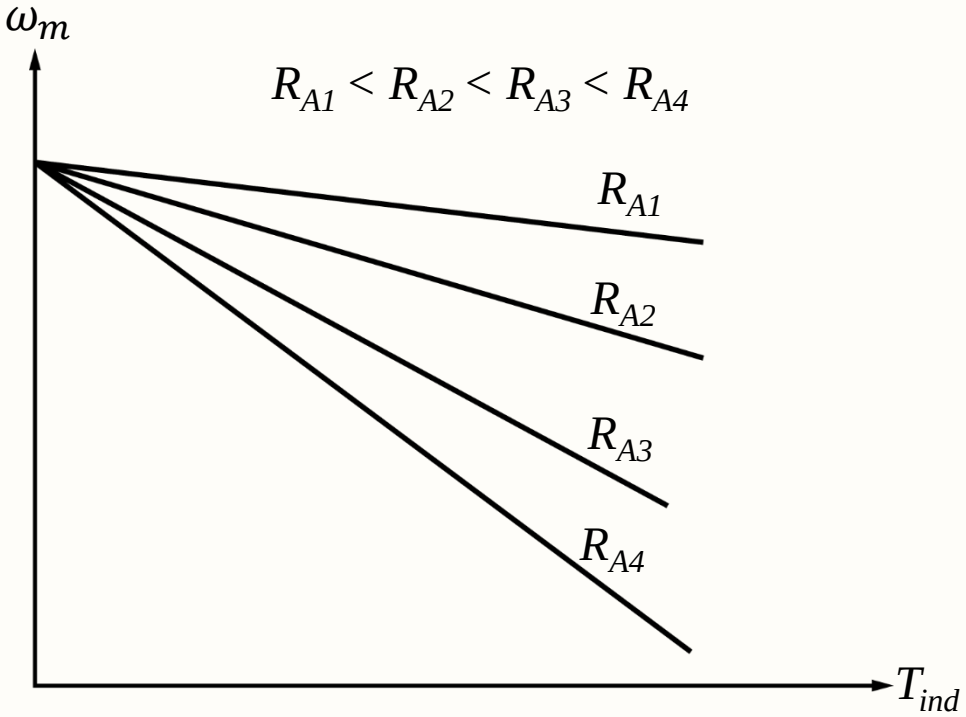
$$\Rightarrow I_A \downarrow \Rightarrow T_{ind} \downarrow \text{ until } T_{ind} = T_{load} \text{ at a higher speed } \omega_m$$



The no-load speed is shifted, but the slope of the curve remains constant.

Speed Control of a Shunt DC Motor

- Inserting a resistor in series with the armature circuit (armature resistance speed control).
  - This method will incur large losses in the inserted resistor.



$$\omega_m = \frac{V_T}{K\phi} - \frac{R_A}{(K\phi)^2} T_{ind}$$

Example 2

A 25-hp, 240-V shunt motor operating at 850 r/min draws a line current of 91 A when operating at rated conditions. A 2.14 ohms resistor inserted in series with the armature causes the speed to drop to 634 r/min. The respective armature-circuit resistance and field-circuit resistance are 0.221 ohm and 120 ohms, respectively. Determine the new armature current.

(Solutions →)

Example 2 – Solutions

Before the addition of the resistor,

$n_{m_1} = 850 \text{ rpm},$

$E_{A_1} = 240 - (91-2)(0.221) = 220.331 \text{ V}$

After the addition of the resistor,

$n_{m_2} = 634 \text{ rpm}, I_{F_1} = I_{F_2} \Rightarrow \text{ flux is constant.}$

$$\frac{E_{A_1}}{E_{A_2}} = \frac{n_{m_1}}{n_{m_2}} \Rightarrow \frac{220.331}{E_{A_2}} = \frac{850}{634}$$

$$\Rightarrow E_{A_2} = 164.3 = 240 - I_{A_2}(0.221 + 2.14)$$

$$\Rightarrow I_{A_2} = 32.06 \text{ A}$$

Speed Control Limits

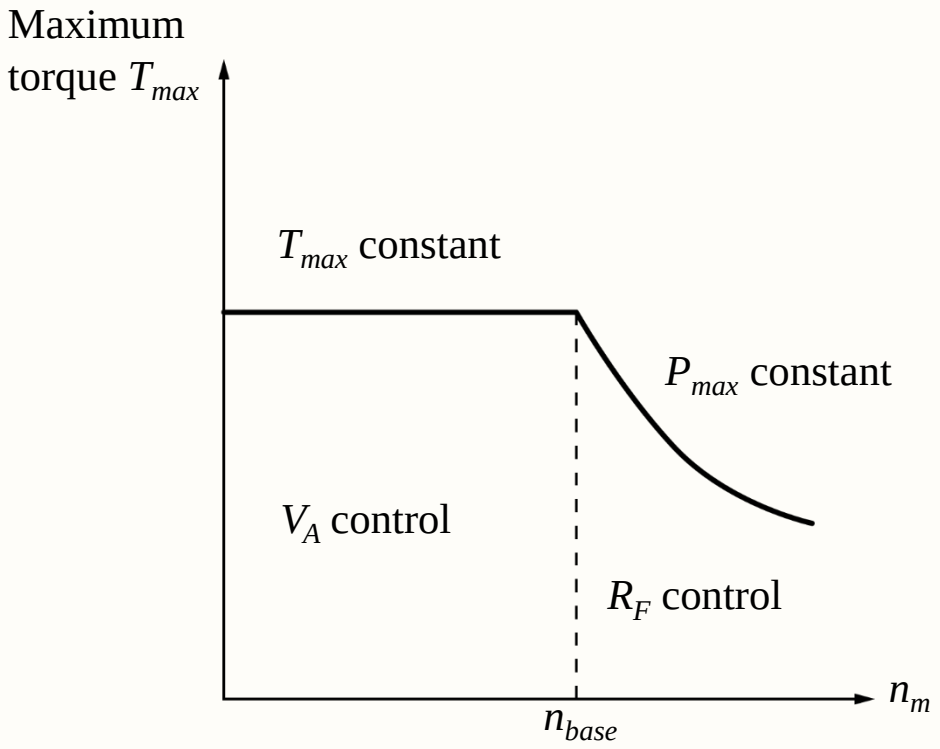
- Field resistance control
  - The lower the field current in shunt (or separately excited) motors, the faster they turn and vice versa.
  - An increase in field current causes a decrease in speed, and hence, there is a minimum achievable speed by field circuit control. (Limited by the maximum permissible field current flowing through it, to avoid burning up the field windings.)
  - It is used to control for speeds above base speed or rated speed.

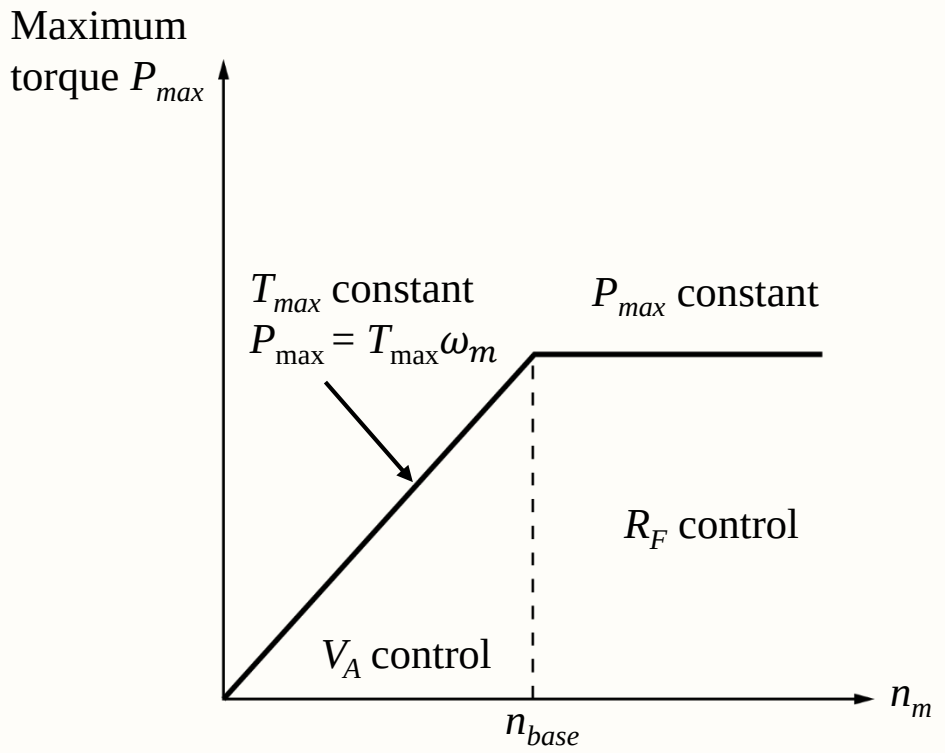
Speed Control Limits

- Armature voltage control
  - The lower the armature voltage (in separately excited motors), the slower they turn and vice versa.
  - An increase in armature voltage causes an increase in speed, and hence, there is a maximum achievable speed by armature voltage control. (Limited by the maximum permissible value of the armature voltage.)
  - It is used to control for speeds below base speed or rated speed. To control speeds above base speed would require excessive armature voltage, possibly damaging the armature circuit.

Speed Control Limits

- The two methods are complementary. Combining the two can get a range of speed variations of up to 40 to 1. Shunt and separately-excited DC motors have excellent speed control characteristics.





Armature Voltage Control

- The flux is constant at rated value as the terminal voltage is adjusted. The maximum torque achievable is independent of the speed of rotation and is given by:

$$T_{\max} = K\phi_{rated} I_{A,\max}$$

- T_{max} is a constant since ϕ_{rated} and the max allowable $I_{A,max}$, without damaging the armature windings, are constants. Conversely, the maximum developed power out of the motor under armature voltage control below rated speed

$$P_{\max} = T_{\max}\omega_m$$

is directly proportional to the operating speed.

Field Resistance Control

- Speed increase is caused by decrease in flux. For speeds beyond the base speeds, the terminal voltage is kept constant at its maximum value, while the field current is adjusted.

- Since $E_A = K\phi\omega_m$ and is relatively constant due to fixed V_T, then ϕ will be directly proportional to $\left(\frac{1}{\omega_m}\right)$. Thus, the maximum torque $T_{max} = K\phi I_{A,max}$ also becomes directly proportional to $\left(\frac{1}{\omega_m}\right)$.

  On the other hand, $P_{max} = T_{max}\omega_m$ remains constant.

Field Resistance Control

- Operation below rated or base speed is known as the constant torque region because the max induced torque is constant.
- Operation above rated speed is known as constant power region.

DC Motor Problems on Starting

- The motor must be protected from physical damage during the starting period.
- For example, a 50-hp, 250-V motor DC motor has an armature resistance of 0.06 ohms. The full-load current is less than 200 A. At starting conditions, the motor is not turning and thus there is no internal voltage generated. The current at starting is:

$$I_A = \frac{V_T - E_A}{R_A} = \frac{250 - 0}{0.06} = 4167 \text{ A}$$

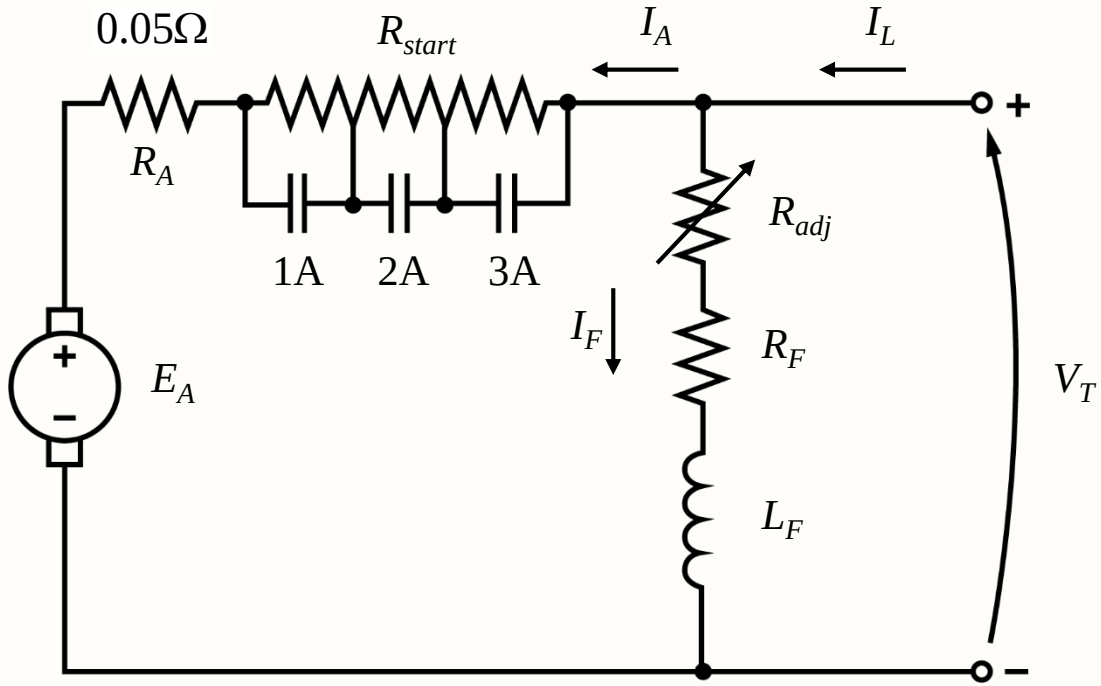
which is more than 20 times the motor's rated current, and this can severely damage the motor.

DC Motor Problems on Starting

- A solution is to insert a starting resistor in series with the armature to limit the current flow, until the internal generated voltage can be build up to do the limiting.
- The resistor must not be in the circuit permanently, because it will result in excessive losses.
- In practice, a starting resistor is made up of a series of pieces, each of which is removed from the motor circuit in succession as the motor speeds up.

Speed Control Limits

- A shunt motor with a starting resistor in series with its armature.
- Contacts 1A, 2A and 3A short-circuit portions of the starting resistor, when they close.



Summary of DC Motors

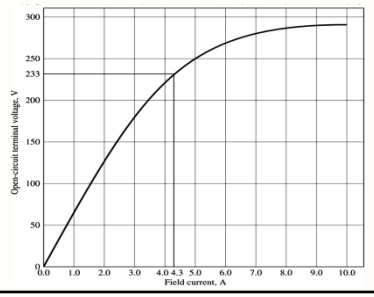
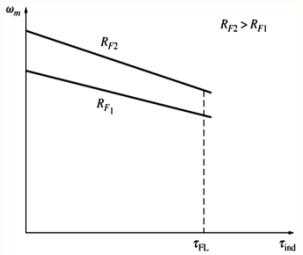
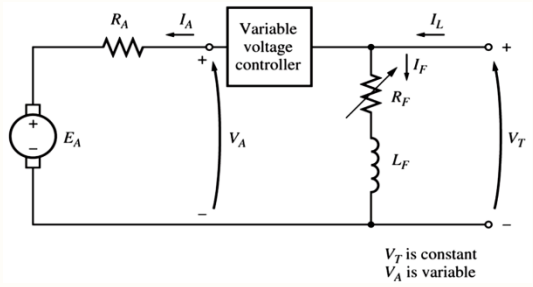
You have learnt:

- There are several types of DC motors, differing in the manner in which their field fluxes are derived.
- A shunt or separately excited DC motor has a torque-speed characteristics whose speed drops linearly with increasing speed. Its speed can be controlled by changing its field current, armature voltage, or armature circuit resistance.
- Combinations of armature voltage and field resistance speed controls allow a wide range of speed variations.

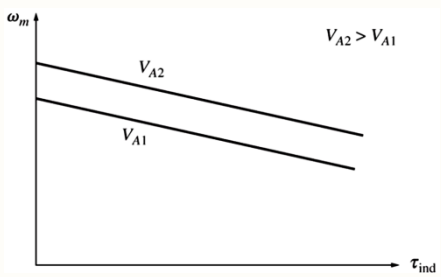
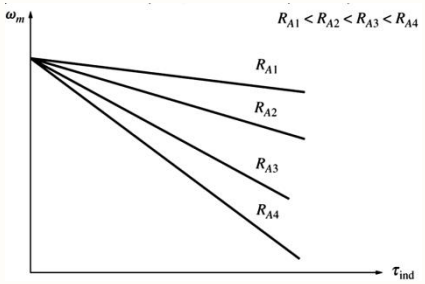
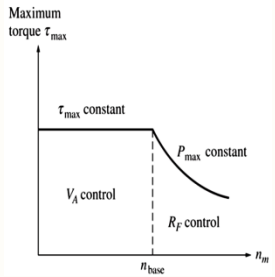
Summary of DC Motors

- The nonlinear analysis of a DC motor.
- The applications of the armature voltage and field resistance speed control methods to allow a wide range of speed variations.
- Starting circuits to limit the starting current to a safe level.

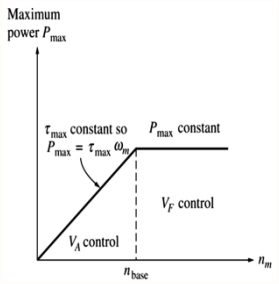
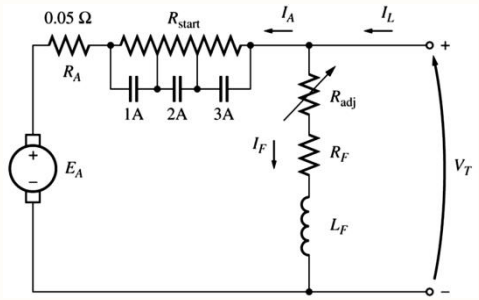
References

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